

Lecture 7 — Eigendecomposition; Symmetric Matrices

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After this lecture you should be able to

- Assemble the eigenvectors into $\mathbf{V}$ and the eigenvalues into $\mathbf{\Lambda}$, and derive $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ from $\mathbf{A}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}$.
- Prove that a real symmetric matrix has real eigenvalues and orthogonal eigenvectors, and state the spectral theorem $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$.
- Compute powers $\mathbf{A}^\ell = \mathbf{V}\mathbf{\Lambda}^\ell\mathbf{V}^{-1}$ and use them to evaluate the matrix exponential.
- Solve the system $d\mathbf{x}/dt = \mathbf{A}\mathbf{x}$ as $\mathbf{x} = e^{\mathbf{A}t}\mathbf{C}$, and read the eigenvectors as the modes of the system.

1 Eigendecomposition

Last lecture ended with a statement in words: in the eigenbasis, $\mathbf{A}$ is diagonal. Today we turn that sentence into a matrix identity.

Let the n linearly independent eigenvectors of $\mathbf{A}$ be $\mathbf{v}_1, \dots, \mathbf{v}_n$, with eigenvalues $\lambda_1, \dots, \lambda_n$. Arrange $\mathbf{v}_1, \dots, \mathbf{v}_n$ column-wise into an $n \times n$ **eigenvector matrix**

$$\mathbf{V} = \begin{bmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{bmatrix}.$$

Now multiply by $\mathbf{A}$ and read the product by columns, as in Lecture 1: column j of $\mathbf{A}\mathbf{V}$ is $\mathbf{A}$ times column j of $\mathbf{V}$. So

$$\mathbf{A}\mathbf{V} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{A}\mathbf{v}_1 & \mathbf{A}\mathbf{v}_2 & \cdots & \mathbf{A}\mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ \lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix},$$

using $\mathbf{A}\mathbf{v}_j = \lambda_j\mathbf{v}_j$ in each column. The last matrix has the same columns as $\mathbf{V}$, each scaled by its own number — and scaling column j by λ_j is exactly what right-multiplication by a diagonal matrix does:

$$\mathbf{A}\mathbf{V} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix} \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}}_{\mathbf{\Lambda}},$$

where $\mathbf{\Lambda}$ is the **diagonal eigenvalue matrix**. We have shown

$$\mathbf{A}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}.$$

Because $\mathbf{V}$ has n linearly independent columns it is invertible, so we may multiply on the right by $\mathbf{V}^{-1}$:

$$\boxed{\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}}$$

the **eigendecomposition** of $\mathbf{A}$.

What the three factors do

Read $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ right to left as a recipe rather than a formula: rewrite $\mathbf{x}$ in the eigenbasis, scale coordinate i by λ_i , change back. Nothing in the middle step mixes coordinates — all of the complication of $\mathbf{A}$ has been pushed into the change of basis, done once at the start and undone once at the end.

$$\mathbf{x} \xrightarrow[\text{rewrite } \mathbf{x} \text{ in the eigenbasis}]{\mathbf{V}^{-1}} \mathbf{V}^{-1}\mathbf{x} \xrightarrow[\text{scale coordinate } i \text{ by } \lambda_i]{\mathbf{\Lambda}} \mathbf{\Lambda}\mathbf{V}^{-1}\mathbf{x} \xrightarrow[\text{go back to the original basis}]{\mathbf{V}} \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}\mathbf{x} = \mathbf{A}\mathbf{x}$$

The example from last lecture

For $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ we found $\lambda_1 = 4$ with $\mathbf{v}_1 = (1, 1)$ and $\lambda_2 = 2$ with $\mathbf{v}_2 = (1, -1)$. So

$$\mathbf{V} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{\Lambda} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{V}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and multiplying the three together does return the matrix we started from:

$$\mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{bmatrix} 4 & 2 \\ 4 & -2 \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = \mathbf{A}.$$

Not every matrix has one

The derivation used invertibility of $\mathbf{V}$, which needed n *linearly independent* eigenvectors. A defective matrix (Lecture 6) does not have them, and for such a matrix $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ simply does not exist — there is nothing to invert. Every formula in the rest of this lecture inherits that hypothesis.

2 Symmetric matrices

Real symmetric matrices — and their extended complex family, the **Hermitian** matrices — are particularly nice, because of their well-behaved eigenvalues and eigenvectors. Since essentially every matrix this course builds from a physical equilibrium turns out to be symmetric, this is the case we care about most.

In particular, if a real-valued $n \times n$ matrix $\mathbf{A}$ is symmetric,

$$\mathbf{A} = \mathbf{A}^T,$$

we have:

- (1) All eigenvalues are **real**.
- (2) There are n eigenvectors, and they are **orthogonal to each other**.

Both facts are worth proving, because both proofs are three lines and both explain *why* symmetry is the hypothesis that makes them work. Throughout we write

$$\mathbf{v}^\dagger \equiv (\mathbf{v}^*)^T$$

for the conjugate transpose — complex conjugate, then transpose. We allow complex vectors here because we do not yet know that λ is real; that is what we are proving.

Proof of (1): the eigenvalues are real

Suppose we have an eigenvalue–eigenvector pair $(\lambda, \mathbf{v})$ for $\mathbf{A}$, so that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. Take the conjugate inner product of both sides with $\mathbf{v}$:

$$\mathbf{v}^\dagger \mathbf{A}\mathbf{v} = \lambda \underbrace{\mathbf{v}^\dagger \mathbf{v}}_{\text{real}}.$$

The quantity $\mathbf{v}^\dagger \mathbf{v} = \sum_i |v_i|^2$ is real and positive, since $\mathbf{v} \neq \mathbf{0}$.

Now $\mathbf{v}^\dagger \mathbf{A}\mathbf{v}$ is just a number (1×1), so conjugating it is the same as taking its conjugate transpose:

$$(\mathbf{v}^\dagger \mathbf{A}\mathbf{v})^* = (\mathbf{v}^\dagger \mathbf{A}\mathbf{v})^\dagger = \mathbf{v}^\dagger \mathbf{A}^\dagger \mathbf{v} = \mathbf{v}^\dagger \mathbf{A}\mathbf{v},$$

the last step because $\mathbf{A}$ is real symmetric, so $\mathbf{A}^\dagger = \mathbf{A}$. A number equal to its own conjugate is real, so $\mathbf{v}^\dagger \mathbf{A}\mathbf{v}$ is real. Thus

$$\lambda = \frac{\mathbf{v}^\dagger \mathbf{A}\mathbf{v}}{\mathbf{v}^\dagger \mathbf{v}}$$

is a real number divided by a real number, and is therefore also real. $\square$

Proof of (2): eigenvectors of distinct eigenvalues are orthogonal

Suppose we have two eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ for a real symmetric matrix $\mathbf{A}$, and $\lambda_1 \neq \lambda_2$. Start from

$$\mathbf{A}\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$$

and take inner products of both sides with $\mathbf{v}_2$:

$$\mathbf{v}_2^\dagger \mathbf{A}\mathbf{v}_1 = \lambda_1 \mathbf{v}_2^\dagger \mathbf{v}_1.$$

Now compute the same left-hand side the other way, moving $\mathbf{A}$ across:

$$\mathbf{v}_2^\dagger \mathbf{A}\mathbf{v}_1 = (\mathbf{A}^\dagger \mathbf{v}_2)^\dagger \mathbf{v}_1 = (\mathbf{A}\mathbf{v}_2)^\dagger \mathbf{v}_1 = \lambda_2 \mathbf{v}_2^\dagger \mathbf{v}_1,$$

again using $\mathbf{A}^\dagger = \mathbf{A}$, and using part (1) at the last step so that $\lambda_2^* = \lambda_2$. Setting the two expressions equal,

$$\lambda_1 \mathbf{v}_2^\dagger \mathbf{v}_1 = \lambda_2 \mathbf{v}_2^\dagger \mathbf{v}_1, \quad \text{with} \quad \lambda_1 \neq \lambda_2.$$

The only way that can hold is

$$\mathbf{v}_2^\dagger \mathbf{v}_1 = 0,$$

and so $\mathbf{v}_1, \mathbf{v}_2$ are orthogonal to each other. $\square$

The repeated case

If $\lambda_1 = \lambda_2$ this proof fails — the last step divided by $\lambda_1 - \lambda_2$. But in that case nothing is lost: any linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ is again an eigenvector for the same eigenvalue, so we can construct combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$ that *are* orthogonal to each other, i.e. construct an orthogonal basis for the degenerate eigenspace. A repeated eigenvalue gives us a whole plane (or higher-dimensional subspace) of eigenvectors, and we get to choose a convenient basis inside it.

Symmetry is the hypothesis, not a bonus

Both proofs used $\mathbf{A}^\dagger = \mathbf{A}$, once each, and would collapse without it. A general matrix can have complex eigenvalues (a rotation has no real eigenvector at all) and its eigenvectors need not be

orthogonal. Do not carry the conclusions of this section over to a non-symmetric matrix.

The spectral theorem

Put (1) and (2) together. Since the n eigenvectors are mutually orthogonal, scale each one to unit length — we are free to, by Lecture 6 — and collect them into $\mathbf{V}$. Then $\mathbf{V}$ has orthonormal columns, which is precisely the statement $\mathbf{V}^\top \mathbf{V} = \mathbf{I}$, i.e.

$$\mathbf{V}^{-1} = \mathbf{V}^\top.$$

The inverse in the eigendecomposition is now free: no elimination, just a transpose.

Spectral theorem

Every real symmetric matrix factors as

$$\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top, \quad \mathbf{\Lambda} \text{ real diagonal}, \quad \mathbf{V}^\top \mathbf{V} = \mathbf{I}.$$

Real eigenvalues, perpendicular eigenvectors, and the change of basis undone by a transpose. This is the single most useful factorisation in the course.

For last lecture's example, normalising $(1, 1)$ and $(1, -1)$ to unit length gives

$$\mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{V}^\top \mathbf{V} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \mathbf{I},$$

so indeed $\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$ with $\mathbf{\Lambda} = \text{diag}(4, 2)$.

3 The matrix exponential

The eigenvector basis for a matrix $\mathbf{A}$ is, in a sense, the best basis for that matrix: it is the basis under which the matrix is diagonal. One particular application of the eigendecomposition is in understanding and computing the **matrix exponential**.

Where it comes from

The scalar first-order linear ODE

$$\frac{dx}{dt} = ax \quad \text{has general solution} \quad x = Ce^{at}.$$

Now take a *system* of first-order linear ODEs, with $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{A}$ an $n \times n$ matrix:

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}.$$

Analogously, the general solution is

$$\mathbf{x} = e^{\mathbf{A}t} \mathbf{C}$$

where $\mathbf{C}$ is now an n -dimensional vector and $e^{\mathbf{A}t}$ is a *matrix exponential*. By the Taylor series expansion of the exponential function,

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{1}{2}(\mathbf{A}t)^2 + \cdots = \sum_{\ell=0}^{\infty} \frac{1}{\ell!} (\mathbf{A}t)^\ell.$$

So the object is defined — but it asks us for every power of $\mathbf{A}$, which looks expensive. The eigendecomposition gives us an easy way to evaluate all these matrix powers.

Powers are free in the eigenbasis

Let $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$. Then

$$\mathbf{A}^2 = \mathbf{V}\mathbf{\Lambda}\underbrace{\mathbf{V}^{-1}\mathbf{V}}_{=\mathbf{I}}\mathbf{\Lambda}\mathbf{V}^{-1} = \mathbf{V}\mathbf{\Lambda}^2\mathbf{V}^{-1},$$

and analogously, for every ℓ ,

$$\mathbf{A}^\ell = \mathbf{V}\mathbf{\Lambda}^\ell\mathbf{V}^{-1}.$$

The interior factors telescope. And $\mathbf{\Lambda}^\ell$ costs nothing: it is the diagonal matrix of λ_i^ℓ .

Key idea

$\mathbf{A}^\ell = \mathbf{V}\mathbf{\Lambda}^\ell\mathbf{V}^{-1}$: to raise a matrix to a power, raise its *eigenvalues* to that power and leave the eigenvectors alone. Anything built out of powers by a series — the exponential above being the case we need — therefore acts on the eigenvalues one at a time.

Assembling the exponential

Substitute the powers into the series:

$$\begin{aligned} e^{\mathbf{A}t} &= \sum_{\ell=0}^{\infty} \frac{1}{\ell!} \mathbf{A}^\ell t^\ell = \sum_{\ell=0}^{\infty} \frac{1}{\ell!} \mathbf{V}(\mathbf{\Lambda}^\ell t^\ell) \mathbf{V}^{-1} \\ &= \mathbf{V} \left(\sum_{\ell=0}^{\infty} \frac{1}{\ell!} \begin{bmatrix} \lambda_1^\ell t^\ell & & \\ & \ddots & \\ & & \lambda_n^\ell t^\ell \end{bmatrix} \right) \mathbf{V}^{-1} \\ &= \mathbf{V} \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix} \mathbf{V}^{-1} = \mathbf{V} e^{\mathbf{\Lambda}t} \mathbf{V}^{-1}. \end{aligned}$$

The sum could be pulled inside because $\mathbf{V}$ and $\mathbf{V}^{-1}$ do not depend on ℓ , and each diagonal entry is then just the ordinary scalar series $\sum_{\ell} \frac{1}{\ell!} (\lambda_i t)^\ell = e^{\lambda_i t}$.

If $\mathbf{A}$ is real symmetric, the spectral theorem replaces the inverse by a transpose:

$$e^{\mathbf{A}t} = \mathbf{V} e^{\mathbf{\Lambda}t} \mathbf{V}^\top.$$

Some physical context

If, for example, $\mathbf{A}$ is the Hamiltonian of a physical system, the eigenvectors $\mathbf{V}$ are referred to as the **modes** of the physical system. The matrix exponential is then related to the normal mode expansion: the system is decomposed into modes, each mode evolves on its own with its own $e^{\lambda_i t}$, and the modes are recombined at the end. That is exactly the three-arrow picture from Section 1, now running in time.

Where this is going

For a symmetric matrix we now know the eigenvalues are real. The next question is whether they are *positive*, and that is Lecture 8: positive definite matrices are the symmetric matrices with all $\lambda_i > 0$. The distinction is not cosmetic — in $\mathbf{x} = e^{\mathbf{A}t}\mathbf{C}$ the sign of each λ_i decides whether that mode grows or decays, so it is the difference between a stable system and an unstable one, and between an energy that is a genuine minimum and one that is not.

The matrix exponential returns in Lecture 9, where $d\mathbf{x}/dt = \mathbf{A}\mathbf{x}$ with $\mathbf{A}$ a second-difference matrix *is* the discretised heat equation, and again in Lecture 14 for oscillations.

Reading

Strang, *Computational Science and Engineering*, §1.5.